

Bicomplex Signal Realization and Fixed-Boundary Point Extensions

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Abstract

We give an explicit unit-power bicomplex state family whose phase-sensitive quadratic readout reproduces the ruled signal map. The required phase coupling is stated as model data. We also prove that three fixed Hermitian quadratic observables on normalized two-component pure states cannot reproduce this whole map. For the actual induced Whitney metric, a singular coordinate change reduces the scalar Dirichlet form to a uniformly elliptic divergence-form problem with bounded density. Interior Hölder estimates then give continuous, graph-bounded values at both interior cross-caps. Consequently the corrected point restriction of the fixed Dirichlet operator has deficiency indices $(2, 2)$, an actual resolvent boundary triple and a $U(2)$ extension family. Its Green vectors are defined by the actual resolvent; explicit logarithmic normalizations and singular Dirac transmission remain separate questions.

1 An explicit signal realization

We use the same completed source labels and map as the verified v13 draft:

$$S(t, u) = (c + 2u(1 - c), (1 - u)s, uQ), \quad c = \cos t, \quad s = \sin t, \quad Q = \sqrt{c(2 - c)},$$

where $-\pi/2 \leq t \leq \pi/2$ and $0 \leq u \leq 1$. The bicomplex idempotent coordinates are $z = z_+e_+ + z_-e_-$; the chosen Hilbert power is $|z_+|^2 + |z_-|^2$. Define the two unit phasors

$$\zeta_+(t) = e^{it/2}, \quad \zeta_-(t) = \sqrt{1 - c/2} + i\sqrt{c/2}.$$

Thus $\zeta_+^2 = c + is$ and $\zeta_-^2 = 1 - c + iQ$. These phases encode the selected orthogonal-circle coupling. Power conservation alone does not select that coupling.

Theorem 1.1 (Quadrature realization). *The continuous state family and fixed real quadratic readout*

$$\begin{aligned} \Phi(t, u) &= \sqrt{1 - u}\zeta_+(t)e_+ + \sqrt{u}\zeta_-(t)e_-, \\ P(z) &= (\Re(z_+^2) + |z_-|^2 + \Re(z_-^2), \Im(z_+^2), \Im(z_-^2)) \end{aligned}$$

satisfy $|\Phi_+|^2 = 1 - u$, $|\Phi_-|^2 = u$ and $P \circ \Phi = S$ on the entire physical source. The lift is smooth where $0 < u < 1$ in the completed source charts.

Proof. Both phasors have modulus one. Their squares give $P\Phi = ((1 - u)c + u(2 - c), (1 - u)s, uQ)$, which is S . The readout measures the real and imaginary quadratures of the component squares, together with one DC power term. It is specified on the ambient algebra before a source point is substituted. At the angular endpoints use $r = Q$, $C = 1 - \sqrt{1 - r^2}$ and $t = \varepsilon \arccos C$. The imaginary component of ζ_- is $r/\sqrt{2(2 - C)}$ for $r \geq 0$ and has a smooth signed extension; its

real component is $\sqrt{1 - C/2}$. The first phasor is smooth in C . This verifies smoothness in the actual endpoint charts. Square-root power weights are smooth in the open ruling interval and continuous at its ends. $\square$

The construction retains a common phase reference: multiplying both components by $e^{i\alpha}$ rotates their squares by $e^{2i\alpha}$. It is a classical phase-sensitive signal readout, rather than a claim about three quantum expectation values.

Proposition 1.2 (Power-coordinate boundary obstruction). *No C^1 lift in the original coordinate u can satisfy $|z_-(t, u)|^2 = u$ up to $u = 0$. Similarly $|z_+(t, u)|^2 = 1 - u$ prevents a C^1 lift at $u = 1$. Writing $u = \sin^2 v$ gives a smooth lift on $0 \leq v \leq \pi/2$, but this substitution is not a boundary diffeomorphism of the original source.*

Proof. At $u = 0$ the component is zero. If it were C^1 , the one-sided derivative of its squared modulus would be $2\Re(\bar{z}_- \partial_u z_-) = 0$, contradicting the derivative of u . The other endpoint is identical. In the power-angle coordinate the weights are $\cos v$ and $\sin v$; the derivative of $\sin^2 v$ vanishes at both ends. $\square$

Proposition 1.3 (Three Hermitian observables do not realize this map). *There are no three fixed Hermitian 2×2 matrices H_j and normalized pure-state family $z(t, u) \in \mathbb{C}^2$ such that $S_j(t, u) = z(t, u)^* H_j z(t, u)$ throughout the physical source.*

Proof. The Bloch vector of a normalized two-component pure state lies on the unit sphere. Expanding H_j in the identity and Pauli matrices gives $P_H(z) = a + Mr(z)$ with a real 3×3 matrix M ; see also the pure-state distinction in [6]. If $\text{rank } M \leq 2$, its image is planar. But the four actual values $(1, 0, 0)$, $(0, 1, 0)$, $(0, -1, 0)$, $(1, 0, 1)$ of S have affine determinant 2. If M is invertible, its image of the sphere is an ellipsoid surface, which contains no nonconstant straight segment. The actual ruling $S(0, u) = (1, 0, u)$ is such a segment. Both cases are impossible. $\square$

The proposition concerns normalized pure states and fixed Hermitian observables. It does not exclude mixed states, additional components, or the phase-sensitive readout already constructed.

2 Resolving the intrinsic scalar metric

For the standard cross-cap $F(x, y) = (x, xy, y^2)$ set $R = \sqrt{x^2 + y^2}$. Its actual pullback metric and a new coordinate map are

$$g_0 = dx^2 + (y dx + x dy)^2 + 4y^2 dy^2, \quad \Psi(x, y) = (X, V) = (x, yR).$$

The coordinate change is singular at zero. It preserves the completed source point and its neighbourhood topology.

Lemma 2.1 (Uniform metric comparison). *The map Ψ is a homeomorphism of $\mathbb{R}^2$ and a smooth diffeomorphism off zero. For $0 < R \leq 1/2$,*

$$\frac{1}{16} \Psi^*(dX^2 + dV^2) \leq g_0 \leq 8 \Psi^*(dX^2 + dV^2).$$

Also $X^2 + V^2 = |F(x, y)|^2$.

Proof. For fixed x , the function $y \mapsto y\sqrt{x^2 + y^2}$ is strictly increasing onto $\mathbb{R}$. Its inverse is continuous and obeys

$$y^2 = \frac{\sqrt{X^4 + 4V^2} - X^2}{2}, \quad \text{sgn } y = \text{sgn } V.$$

Off zero the Jacobian is $(x^2 + 2y^2)/R > 0$. Continuity at zero follows from the inverse formula. The radius identity follows by expansion. For a tangent vector (ξ, η) put $h = \xi^2 + R^2\eta^2$ and $E = g_0((\xi, \eta), (\xi, \eta))$. The exact positive decompositions

$$\begin{aligned} E - \tfrac{1}{2}h &= \tfrac{1}{2}(2y\xi + x\eta)^2 + (\tfrac{1}{2} - y^2)\xi^2 + \tfrac{7}{2}y^2\eta^2, \\ 4h - E &= (y\xi - x\eta)^2 + (3 - 2y^2)\xi^2 + 2x^2\eta^2 \end{aligned}$$

give $h/2 \leq E \leq 4h$. Now $dV = a dx + b dy$, where $a = xy/R$, $b = (x^2 + 2y^2)/R$. Here $|a| \leq R/2 \leq 1/4$ and $1 \leq b/R \leq 2$. Consequently $\xi^2 + (a\xi + b\eta)^2 \leq 8h$. Conversely $R\eta = (R/b)((a\xi + b\eta) - a\xi)$ gives $h \leq 2(\xi^2 + (a\xi + b\eta)^2)$. Combining these estimates proves the displayed comparison. $\square$

Grieser already proved that Whitney-umbrella metrics become quasi-Euclidean under a singular coordinate change [1, Section 1]. We give the explicit comparison here to track its scalar PDE hypotheses; the coordinate method is not claimed as a new discovery.

Corollary 2.2 (Application to the actual ruled germs). *Each of the actual interior germs $P_\pm$ admits a source homeomorphism, smooth off its singular label, in which its metric tensor G and density $m = \sqrt{\det G}$ are bounded above and below by positive constants. The divergence matrix $a = mG^{-1}$ is bounded, measurable and uniformly elliptic.*

Proof. The verified adapted determinants in v13 show that each actual germ is a smooth cross-cap. Thus an actual source diffeomorphism h and an ambient target diffeomorphism L give $S = L \circ F \circ h$, after translations. Shrink the neighbourhood so the singular values of DL lie between positive constants ℓ and M . Then $\ell^2 g_0 \leq F^*(L^* g_{\mathbb{R}^3}) \leq M^2 g_0$. Apply Lemma 2.1 to $\Psi \circ h$. Bounds on the eigenvalues of G imply the asserted density and divergence-matrix bounds. This is metric comparison through an actual target map, not an assertion that L is an isometry or that it preserves sheet parity. $\square$

3 Actual graph-bounded point traces

Let $\mathcal{H} = L^2(D_{\text{reg}}, dA_g)$ and let q_D be the Friedrichs form of the actual scalar metric, with the regular outer Dirichlet boundary fixed. Its self-adjoint operator is H_F . The all-five-point capacity proof, compact form embedding and positive Dirichlet gap are the proved v13 inputs. Only the two interior labels are used for the point restriction below.

Theorem 3.1 (Actual point evaluation). *Every $f \in \text{Dom } H_F$ has a unique continuous representative near $P_\pm$ in the completed source. In the resolved coordinates it is locally Hölder continuous, and*

$$\tau f = (f(P_+), f(P_-)), \quad \|\tau f\| \leq C(\|f\|_{\mathcal{H}} + \|H_F f\|_{\mathcal{H}}).$$

The trace $\tau : \text{Dom } H_F \rightarrow \mathbb{C}^2$ is surjective and its kernel is dense in $\mathcal{H}$.

Proof. On a resolved coordinate disk the form norm is equivalent to H^1 and the Hilbert norm to ordinary L^2 , by Corollary 2.2. A point has zero H^1 capacity in dimension two: cutoffs on a logarithmic annulus have energy tending to zero. Thus the local completed form space is H^1 on the whole disk. This also follows by localizing the original Friedrichs core and completing it in the equivalent norm. Writing $b = H_F f$, its weak equation becomes

$$-\text{div}(a\nabla f) = mb.$$

It holds across zero. Indeed tests on the whole disk can be approximated in H^1 by tests avoiding zero, and both the form pairing and the L^2 right-hand side are continuous in that norm. There is no unaccounted delta term in this equation for a Friedrichs-domain function.

The coefficients are real symmetric, measurable and uniformly elliptic; m is bounded; and $mb \in L^2$, with $2 > \dim(D)/2 = 1$. The interior inhomogeneous De Giorgi–Nash estimate gives, on a smaller disk,

$$\|f\|_{C^\alpha} \leq C(\|f\|_{L^2} + \|mb\|_{L^2}).$$

This is the standard external scalar theorem, for example [2, Theorem 8.24]; the homogeneous estimate and L^p , $p > n/2$, inhomogeneous version are also stated in [3, Questions 52,58]. For a local formulation one can subtract the zero-boundary solution of the inhomogeneous equation on the disk and apply the homogeneous estimate to the remainder. Apply the theorem to real and imaginary parts separately. This yields a continuous value and the claimed graph estimate. Two continuous representatives equal almost everywhere must agree, since the resolved density is bounded below. The coordinate homeomorphism then supplies a unique value at the actual source label.

Choose two disjoint source cutoffs equal to one near the respective point and zero near all outer boundaries and the other singularities. Capacity zero puts them in the form domain; their Laplacians are smooth and supported on regular annuli, so they are in $\text{Dom } H_F$. Their traces are the two standard basis vectors, proving surjectivity. Finally the original punctured interior smooth compact core lies in $\ker \tau$ and is dense in $\mathcal{H}$. $\square$

Corollary 3.2 (The corrected two-point operator). *The actual operator*

$$A = H_F|_{\ker \tau}$$

is closed, densely defined, symmetric, strictly positive and has deficiency indices $(2, 2)$. Its Friedrichs extension is H_F and all its self-adjoint extensions are parametrized by $U(2)$.

Proof. Graph continuity makes $\ker \tau$ graph closed. The remaining basic properties are inherited from H_F and Theorem 3.1. For nonreal z , $\tau(H_F - z)^{-1}$ is a bounded surjection to $\mathbb{C}^2$ with kernel $\text{ran}(A - z)$. Hence this closed range has codimension two, so $\dim \ker(A^* - \bar{z}) = 2$. Von Neumann’s unitary classification applies. The punctured form core is contained in $\text{Dom } A \subset \text{Dom } H_F$; its form closure is exactly q_D . Therefore the Friedrichs extension is H_F . $\square$

This construction is the fixed-boundary point restriction used in regular point-interaction theory, e.g. [4], with its graph-trace hypothesis now verified for the actual Whitney metric. The old graph closure of the interior compact core remains a different operator with infinitely many outer-boundary deficiency modes; its historical $(2, 2)$ statement stays false.

4 Actual Green vectors and boundary pairing

The point traces allow an exhaustive domain description without guessing a logarithmic coefficient. Fix a real $\eta < 0$ and set

$$R_z = (H_F - z)^{-1}, \quad G_z = [\tau R_{\bar{z}}]^* : \mathbb{C}^2 \rightarrow \mathcal{H}.$$

All adjoints use a conjugate-linear first inner product.

Theorem 4.1 (Resolvent boundary triple). *The actual adjoint domain decomposes as*

$$\text{Dom } A^* = \text{Dom } H_F \dot{+} G_\eta \mathbb{C}^2, \quad A^*(u + G_\eta a) = H_F u + \eta G_\eta a.$$

With $\Gamma_0(u + G_\eta a) = a$ and $\Gamma_1(u + G_\eta a) = \tau u$,

$$\langle A^* f, g \rangle - \langle f, A^* g \rangle = \langle \Gamma_1 f, \Gamma_0 g \rangle - \langle \Gamma_0 f, \Gamma_1 g \rangle.$$

The pair (Γ_0, Γ_1) is surjective onto $\mathbb{C}^2 \oplus \mathbb{C}^2$. Every self-adjoint extension is specified by

$$(U - I)\Gamma_1 f + i(U + I)\Gamma_0 f = 0, \quad U \in U(2).$$

In this convention $U = I$ gives H_F .

Proof. The bounded map $T_\eta = \tau R_\eta$ is onto; its adjoint is injective and has range $\text{ran}(A - \eta)^\perp = \ker(A^* - \eta)$. For $f \in \text{Dom } A^*$ put $u = R_\eta(A^* - \eta)f$. Then $f - u$ belongs to that kernel. The intersection with $\text{Dom } H_F$ is zero since η is in the resolvent set of H_F . This proves the unique decomposition and action. For $f = u + G_\eta a$, $g = v + G_\eta b$, the u, v and a, b terms cancel. The cross terms use

$$\langle (H_F - \eta)u, G_\eta b \rangle = \langle \tau u, b \rangle$$

and its conjugate, giving the stated Green form. Given (a, b) , take u with $\tau u = b$ and set $f = u + G_\eta a$. The kernel of both traces is $\text{Dom } A$. Self-adjoint restrictions correspond to maximal isotropic planes of this finite nondegenerate Green form. The Cayley transform gives exactly the displayed unitary equation, including planes that are not graphs of a Hermitian matrix. $\square$

Corollary 4.2 (Actual finite-rank resolvent formula). *Let $B = B^*$ and impose $\Gamma_1 = B\Gamma_0$. With*

$$M(z) = (z - \eta)\tau R_z G_\eta,$$

the resolvent, wherever both sides are defined, is

$$(H_B - z)^{-1} = R_z + G_z(B - M(z))^{-1}G_z^*.$$

Thus the difference from the fixed Dirichlet resolvent has rank at most two.

Proof. The resolvent identity gives $G_z = G_\eta + (z - \eta)R_z G_\eta$; in particular $\Gamma_0 G_z a = a$ and $\Gamma_1 G_z a = M(z)a$. A solution of $(A^* - z)f = h$ has the unique form $f = R_z h + G_z a$. Its boundary condition becomes $(B - M(z))a = \tau R_z h = G_z^* h$. Solving this finite system gives the formula. $\square$

Corollary 4.3 (Compact extensions and the actual Krein kernel). *Every self-adjoint extension of A has compact resolvent. Its Krein-von Neumann extension A_K has a two-dimensional kernel and satisfies*

$$\text{Dom } A_K = \text{Dom } A \dot{+} G_0 \mathbb{C}^2, \quad \Gamma_1 f = M(0)\Gamma_0 f.$$

Proof. For nonreal z , the difference of the resolvents of any extension and H_F has range in $\ker(A^* - z) = G_z \mathbb{C}^2$, since both solve the same adjoint equation. Hence this difference has rank at most two. The already proved compactness of R_z gives compactness of every extension resolvent. Strict positivity and the positive-operator Krein theorem [5, Theorem 2.1, equation (2.8)] give $\text{Dom } A_K = \text{Dom } A \dot{+} \ker A^*$. Because 0 belongs to the resolvent set of H_F , the same trace-resolvent argument at zero yields $\ker A^* = G_0 \mathbb{C}^2$, of dimension two. Finally $\Gamma_1 G_0 = M(0)\Gamma_0 G_0$ and both traces vanish on $\text{Dom } A$, which identifies its boundary plane. $\square$

This is the Krein extension of the corrected point restriction A . The old compact-core operator still has an infinite-dimensional Krein kernel.

The vectors $G_z e_\pm$ are actual graph-dual point-source Green vectors. The theorem excludes additional defect directions for this specified operator A ; it does not identify their coefficients with those of a particular $\log \rho$ ansatz. Changing the reference η changes the regular boundary coordinate and Weyl normalization, not the actual extension family. An explicit local logarithmic expansion remains useful for geometric interpretation and quantitative couplings.

5 What this establishes for the Dirac problem

The scalar theorem does not require derivatives of the resolved coefficient matrix. A first-order spin operator does, through its connection, so its domain cannot be inferred from scalar uniform ellipticity.

Proposition 5.1 (Actual curvature integrability). *At an actual smooth cross-cap germ, the Gaussian curvature belongs locally to $L^p(dA_g)$ for $1 \leq p < 3/2$. On the standard germ,*

$$K = -\frac{4y^2}{(x^2 + 4y^2 + 4y^4)^2}.$$

In a source disk of radius ε , the absolute curvature integral is at most $C\varepsilon$.

Proof. For the actual map in smooth adapted source coordinates, all second derivatives are bounded. The coefficients of its second fundamental form are consequently bounded, because the unit normal has length one. Metric comparison with the standard germ gives $\det g \asymp R^2$ and $dA_g \leq CR dx dy$. The Gauss formula therefore gives $|K| \leq C/R^2$. Polar integration bounds the p th curvature integral by $C \int_0^\varepsilon R^{2-2p} dR$, finite precisely in the stated sufficient range. For $p = 1$ this is $C\varepsilon$. For the standard map direct first- and second-fundamental-form calculation gives the displayed exact formula. $\square$

This supplies an actual integrable spin-curvature input: the curvature of the two-dimensional spin connection is proportional to $K dA_g$, and tensoring by the flat observation sign line adds no local curvature. It does not by itself provide a bounded conformal factor, a controlled connection gauge, or the weighted graph-domain comparison needed at the singular cut. Those estimates, the phase-line fibre identification and a Clifford-compatible transmission map remain the next Dirac obligations. The one-channel-per-chirality exact-cone calculation retains its model scope.

6 Evidence and relation to the verified checkpoint

Result	Evidence and scope
Signal realization	Explicit global continuous lift, interior/completed-chart smoothness and endpoint obstruction; quadratic identities replayed exactly.
Metric comparison	Written homeomorphism and positive estimates; standard metric energies checked independently and in Lean. Grieser’s earlier coordinate result is credited.
Point traces	Actual metric comparison and form-domain argument, followed by the external scalar Hölder theorem; not Lean formalized.
Extension domains	Written Hilbert-space proof for $A = H_F _{\ker \tau}$, actual Green vectors and finite-rank formula; no claimed logarithmic parametrix.
Dirac progress	Actual curvature bound and exact standard curvature formula; singular spin-domain and transmission classification remain open.

This continuation adds results to the locally verified v13/companion 0.01 checkpoint. Its historical PDF, sealed package, manifests and receipts are unchanged. Exact algebra and the listed Lean theorems do not certify the external PDE regularity theorem or the full operator-domain argument. The explicit signal family is new supplied model data, not recovered historical source. Novelty of the complete application requires further literature comparison; no physical capacity, particle or chirality prediction is established by these results.

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